K-Nearest Neighbors (KNN) - Manual Calculation (K=5)

Given:

Test Data: Height = 161 cm, Weight = 61 kg
Mean Height = 164.44, Std Height = 4.36
Mean Weight = 62.11, Std Weight = 2.83

Step 1: Standardization (Z-score)

z_h = (161 - 164.44) / 4.36 = -0.79
z_w = (61 - 62.11) / 2.83 = -0.39

Step 2: Euclidean Distance Calculation

No	Height	Weight	Size	Distance
1	158	58	M	1.24
2	158	59	M	0.98
3	158	63	M	0.91
4	160	59	M	0.75
5	160	60	M	0.43
6	163	60	M	0.58
7	163	61	M	0.46
8	160	64	L	1.11
9	163	64	L	1.13
10	165	61	L	0.92
11	165	62	L	0.98
12	165	65	L	1.62
13	168	62	L	1.66
14	168	63	L	1.75
15	168	66	L	2.4
16	170	63	L	2.11
17	170	64	L	2.29
18	170	68	L	3.26

Step 3: Sort distances and select K=5 nearest neighbors

Nearest 5: 0.43(M), 0.46(M), 0.58(M), 0.75(M), 0.91(M)
Majority Vote -> M = 5, L = 0

Final Prediction: Size = M